

Hey, everybody. Welcome to your Chapter 21 review. In this episode, we'll look closely at one of those special features you'll only get with this program, the Optimum Performance Training model, also known as the OPT model. Think about all the variables that go into program design, especially early in your career. It's OK for things to feel a bit overwhelming. It's a complicated process with a lot to consider, including a client's goals, physical abilities, fitness level, and medical history. And then on top of that, programs need individuality and uniqueness to drive commitment and adherence.

Thankfully, NASM offers a solid solution with the OPT model. It's an integrated, multi-component approach to program design that includes flexibility, cardiorespiratory, core, balance, plyometric SAQ, and resistance training. The OPT model was developed to improve physical abilities using a systematic, progressive, and holistic approach. It's proven to be successful in helping clients from diverse populations reduce their body fat, increase lean muscle mass, improve athletic performance and overall health. And better yet, it's an evidence-based model, meaning it's been tested and validated in lab settings and with real-life human subjects. So you can breathe easy.

Once you're done with this chapter, you should be able to identify the differences between linear and undulating periodization; the differences between macrocycles, mesocycles, and microcycles; evaluate exercise protocols used within all five phases of the OPT model; and use the model to design fitness programs for various client goals. So why is the concept of the OPT model so important? Because doing things in a safe and effective progression just makes sense.

We all want exercise to take place at regularly planned intervals of time, which will vary stress placed on the body to produce results and prevent injury. A training plan clarifies what forms of training will be used, how long it will take, how often it will change, and what specific exercises and techniques will be performed. It's your roadmap to achieving results.

Fitness and sports performance programming is based on the concept of periodization. An annual plan or macro cycle shows how the program will progress from month to month over a one year period. The annual plan is further broken down into specific periods of training. Monthly cycles are called meso cycles, and weekly plans are called micro cycles.

Periodization has shown to be an effective method of program design for many fitness-related goals. It provides for the repeated use of different forms of training at specific times to elicit different adaptations in the body, such as stabilization, muscle endurance, strength, and power. It also manages training more effectively to help reduce the risk of injury and overtraining. There are two types of periodization to utilize-- linear periodization, which gradually increases the intensity of the training load while simultaneously decreasing volume over a set period of time, typically month over month; and undulating periodization, which uses changes in volume, intensity, and exercise selection to provide loading differences on a daily or weekly basis.

NASM designed the OPT model as a basis to program design for clients with various goals and abilities. It uses a workout template to organize daily exercise routines into six parts-- warmup, activation, skill development, resistance training, client's choice, and cool down. This sequence offers structure, as well as flexibility, to tailor the program to your client's specific goals and preferences.

The OPT model further streamlines the periods of training into three levels-- stabilization, strength, and power. The first level, stabilization, is crucial, not only for beginners, but most other populations, too. It's designed to prepare the body for the demands of higher levels of training, although it's also a useful recovery level for advanced clients performing more intense bouts of training. Stabilization involves relatively lower intensity, higher repetition programming, and emphasizes the improvement of movement patterns, core and joint stabilization, muscular endurance, and mobility.

Once a client demonstrates movement competency, the exercises progress to challenge proprioception and postural control. This level has just the one phase of training, phase one stabilization endurance training, and focuses on acquiring and mastering basic exercise skills. The second level, strength, is intended to increase strength endurance, muscle size, and maximal strength. It's comprised of three phases.

Phase two, strength endurance training, utilizes superset techniques. This involves a more strength-focused exercise, immediately followed by a stabilization-focused exercise with similar biomechanical motions. Phase three, muscular development training, stresses the body to achieve maximal hypertrophy. Here, you're focusing on high levels of training volume.

And lastly, phase four, maximal strength training, focuses on increasing the mechanical loads placed on the tissues of the body to develop maximal strength capabilities. Not all clients will need or want to do everything in this strength level. Be sure you understand your client's unique set of circumstances, and weigh it against the various benefits of moving through the different phases in this level of work.

The third level of training, power, is pretty advanced and is designed to increase rate of force production. It's been shown that to develop optimal levels of power, you must train with both heavy loads and light loads at high speeds. The power level has just one phase, phase five, power training, which is all about high force and velocity to increase power. And again, it's not necessarily for everybody, but it's a huge benefit to have a structure that supports literally every level.

Applying the OPT model will be based on the specific goals of your client, whether it's feeling better, fat loss, increase lean body mass, or general performance in sport or in life. If your client wants to reduce body fat, they'll need to expend more calories than they consume. And the best way to do this is to increase the calories burned through structured exercise and daily physical activity. Since the goal isn't dependent on maximal strength or power, they likely only need to be cycled through the first three phases combined with cardiorespiratory training for additional calorie expenditure and overall health and fitness improvements.

For clients looking to increase lean body mass, higher volumes of training will need to be performed to increase muscle size. This client type can be cycled through the first four phases of the OPT model. Depending on their specific needs and wants, cardio training can be included for this kind of client, as well, though, in this case, it's to ensure the cardiorespiratory system is efficient and promoting optimal tissue recoverability.

For improving general performance, you'll need to build a program that increases overall movement control, proprioception, strength, and rate of force production. This type of training is exciting because it can use the entire OPT model. That being said, for the typical client, phases one, two, and five will be the most important.

They'll begin in phase one to gain proper muscle balance and endurance before progressing to phase two and cycling back again between these two phases. Then strategically, you will introduce phase five with an undulating periodization scheme. And again, you will want to include cardio training for all the additional benefits it provides.

The OPT model gives you a framework to build exciting programs with an integrated, multi-component approach to design. It enables you to consider each client's goals, personal preferences, and also, take into account the needs uncovered by their assessment results in order to make the exercise programs impactful and generate long-term adherence. For variety and optimal results, you will combine flexibility, cardiorespiratory, core, balance, plyometric SAQ, and resistance training exercises to minimize boredom and prevent plateauing.

By focusing on teaching proper movement patterns with the correct progression, you'll help your clients improve their mobility and stability while enhancing their endurance and reducing the risk of injury. As it relates to the NASM training methodology, this is arguably the most important chapter in the text. It really helps you, as an aspiring CPT, to understand the specific strategies to design and implement effective training programs.

Essentially, the OPT model utilizes all of the information you learned from the previous chapters and combines it to form a progressive and systematic approach to programming and training. And hopefully, it helped create a clearer picture about what your career can look like and how fun it will be to take all this work you've done and put it into practice. You're in the homestretch now, so keep up the amazing work. And as always, wishing you the very best in your studies and your life.